New timeline proposal - Thesis Alvin

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Previous Organization

Timeline: From the 2nd of march 2026 to the 5th of October 2026
Schedule: Full time, with one month break in July.
Intermediate presentation: 04.06.2026 from 15:00 to 16:00
Master exam: 26.10.2026 from 13:00 to 14:00

New Organization

Timeline: From the 13th of April 2026 to the 16th of November 2026
Schedule: Full time, with one month break in July.
Intermediate presentation: TBD in meeting
Master exam: TBD in meeting

Intermediate steps - same as before

- Review literature on statistical methods for analysing ranking and score data.
- Describe, implement, reproduce and compare the methods identified in the literature review. To this end, the data from Gallo et al. (2023) will be analysed. This will include at minimum 4 methods:
    1. Analysing scoring data using means and standard deviations (Harman, 1998).
    2. Analysing scoring data using hierarchical Bayesian models (Heyard et al., 2022).
    3. Analysing ranking data using the Mallows model (Mallows, 1957).
    4. Jointly analysing score and ranking data using the Mallows-Binomial model (Pearce & Erosheva, 2022).
- Describe the methods used in the JAGS and INLA `R` packages.
- Translate the code and functions from the `ERforResearch` package from a JAGS implementation to a INLA implementation (Heyard et al., 2022).
- Describe and implement a Bayesian imputation model to fill incomplete ranking data.
- Define and implement a joint modelling method for ranks and scores using the INLA package (if time allows).
- Compare the results of the models identified in the literature review and the joint model (if time allows).

Previous deliveries timeline - march start

The deliveries are expected at the first meeting of each month.

1. **April:** Draft of literature review, description of method (1) and (2) and implementation thereof on the Gallo et al. (2023) data.
2. **May:** Literature review, description, implementation and comparison of methods (1) to (4).
3. **June:** Description of Markov Chain Monte Carlo (MCMC) and integrated nested Laplace (INLA) methods for posterior likelihood approximation.
4. **July:** `ERforResearch` package translated to the INLA framework.
5. **September:** Description and implementation of Bayesian imputation methods for ranking data.
6. **October:** Full thesis submission.

New deliveries timeline - mid-april start

The deliveries are expected at the second meeting of each month.

1. **May:** Draft of literature review, description of method (1) and (2) and implementation thereof on the Gallo et al. (2023) data.
2. **June:** Literature review, description, implementation and comparison of methods (1) to (4).
3. **July:** Description of Markov Chain Monte Carlo (MCMC) and integrated nested Laplace (INLA) methods for posterior likelihood approximation.
4. **September:** `ERforResearch` package translated to the INLA framework.
5. **October:** Description and implementation of Bayesian imputation methods for ranking data.
6. **November:** Full thesis submission.

References

Harman, G. (1998). The Management of Quality Assurance: A Review of International Practice. *Higher Education Quarterly*, *52*(4), 345–364. https://doi.org/10.1111/1468-2273.00104 (cit. on p. 1).

Heyard, R., Ott, M., Salanti, G., & Egger, M. (2022). Rethinking the Funding Line at the Swiss National Science Foundation: Bayesian Ranking and Lottery. *Statistics and Public Policy*, *9*(1), 110–121. https://doi.org/10.1080/2330443X.2022.2086190 (cit. on p. 1).

Mallows, C. L. (1957). Non-null ranking models. i [Publisher: [Oxford University Press, Biometrika Trust]]. *Biometrika*, *44*(1), 114–130. https://doi.org/10.2307/2333244 (cit. on p. 1).

Pearce, M., & Erosheva, E. A. (2022). A unified statistical learning model for rankings and scores with application to grant panel review. https://arxiv.org/abs/2201.02539 (cit. on p. 1).